Adaptive Retransmission Scheduling in Underwater Acoustic Sensor Networks based on ARQ_HMAC

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Abstract

Underwater Acoustic Sensor Networks (UASNs) are used in many areas such as ocean monitoring, environmental studies, underwater exploration, and surveillance. Unlike terrestrial wireless networks that depend on radio waves, underwater communication is carried out mostly through acoustic signals. Since acoustic waves travel very slowly and the available bandwidth is quite narrow, communication tends to suffer from large delays [2, 3, 4, 5]. Because of this, building a reliable and efficient underwater communication system is far from easy. A common problem in UASNs is packet collision, which happens when multiple nodes try to access the channel at the same time. When collisions occur, packets fail to reach the destination and must be retransmitted. Retransmissions take more time, raise the chance of further collisions, and waste a significant amount of energy [3, 5, 9]. In addition, due to the long propagation times involved, retransmission schemes designed for terrestrial systems do not always translate well to the underwater setting [2, 8]. The proposed research work is an extension of the ARQ_HMAC protocol that has been specifically designed for UASNs [1]. It uses a contention based channel access along with ARQ, where retransmissions are performed using TDMA [1]. In the case of normal operation, nodes compete for the channel, but when

a transmission fails, the node estimates the propagation delay and reschedules the retransmission in a slot different from the one used initially [1, 8]. A notable drawback of this retransmission strategy is that the propagation delay is estimated only once, at network setup [1]. In reality, underwater conditions keep changing even after the network is deployed, which means the original estimate can become inaccurate fairly quickly [3, 7]. Several studies have shown that underwater channels and propagation characteristics are highly dynamic, so an adaptive retransmission mechanism becomes necessary to cope with such variations [7, 3]. When the estimated delay no longer matches the actual value, it can cause poorly timed retransmissions, more collisions, longer delays, and extra retransmission attempts. The goal of this work is to redesign the retransmission mechanism so that it adapts to changing channel conditions instead of depending on a one-time RTT estimate. The proposed method will keep updating the RTT based on observed network behavior and will trigger retransmissions with the latest value whenever a significant change is detected. Both the original ARQ_HMAC and the proposed adaptive version will be implemented on the UnetStack simulator, and their performance will be compared using Packet Delivery Ratio (PDR), throughput, number of retransmissions, collision rate, and end-to-end delay [1, 7].

1 Introduction

1.1 Background

Underwater Acoustic Sensor Networks (UASNs) have become an important technology for a wide variety of applications, ranging from oceanographic data collection and environmental monitoring to offshore exploration, disaster warning systems, and defense operations [2, 3]. Unlike terrestrial wireless sensor networks that use radio-frequency (RF) signals, UASNs rely on acoustic waves because RF signals get absorbed very quickly in water and optical signals scatter heavily in such environments [2, 4]. However, the acoustic channel behaves very differently from an RF channel. Sound travels in water at around 1500 m/s, which is five orders of magnitude slower than electromagnetic waves in air. As a result, end-to-end delays can range anywhere from tens of milliseconds to several seconds, depending on how far apart the nodes are [2, 5]. The usable bandwidth is also very limited, often just a few kilohertz, and it shrinks further as transmission distance grows [3, 4]. The constraints mentioned above make the task of designing protocols for UASNs much harder than that of designing protocols for normal wireless networks [2, 5]. Here, the importance of MAC layer can be seen clearly since it is the layer that determines the way by which many nodes access the same acoustic channel [4, 5].

1.2 Challenges in Underwater Acoustic Sensor Networks

Designing MAC and retransmission protocols for UASNs comes with several difficulties, most of which stem from the harsh and ever-changing nature of the underwater acoustic environment. The biggest issue is the long and variable propagation delay, which makes it hard to directly adopt existing terrestrial MAC or ARQ schemes [2, 8]. Because signals take so long to travel, there is a large uncertainty window during which a sender cannot really know what is happening at the receiver's end [3, 5]. This greatly increases the chance of collisions, where packets arriving at the same time at a receiver interfere with each other [3]. Once a collision occurs, retransmissions are needed, which drains more energy, lengthens delays, and adds further pressure on the already contested channel [5, 9]. The narrow bandwidth makes this situation worse, since every retransmission eats up a significant share of the available capacity [2, 4]. On top of this, underwater channels change constantly due to shifting temperature, salinity, currents, surface reflections, and movement of nodes or marine life [3, 7]. These environmental factors cause the sound speed, multipath patterns, and attenuation to vary over time, which directly impacts the propagation delay and link quality [3, 7]. Energy is another serious concern, since underwater nodes typically run on batteries that are hard to replace or recharge once deployed, which makes energy-aware design a must [5, 9].

1.3 Existing Techniques and Their Constraints

To cope with these problems, many MAC protocols have been designed for UASNs that could be categorized into contention free, contention-based, and hybrid classes [4, 5]. Contention-free protocols like TDMA and FDMA give each node a dedicated resource, but they suffer from poor channel utilization and need very tight synchronization, which becomes problematic given the large delays underwater [2, 4]. Contention-based protocols such as ALOHA and CSMA variants are more flexible and avoid strict synchronization, but they tend to collide heavily when traffic grows [4, 5]. Hybrid protocols try to combine the strengths of both by switching between the two modes based on network state [1, 4]. A more recent example is ARQ_HMAC, which pairs a contention-based CSMA phase with a TDMA-based retransmission phase to reduce collisions during retransmission attempts [1]. However, most of these protocols, including ARQ_HMAC, depend on a static or one-time propagation delay estimate taken during initialization, which can become inaccurate as the underwater environment keeps evolving [1, 7].

1.4 Problem Statement

In the ARQ_HMAC protocol, the round trip delay estimation is done just one time during the setup phase of communication. This static value is utilized throughout all subsequent communication for making decisions on retransmission scheduling. However, it is worth mentioning that the channel conditions in an underwater environment are subject to changes, and thus the true round trip delay can deviate from the estimated value dramatically.

1.5 Objectives of the Proposed Work

The three main objectives:

1. Design an adaptive retransmission scheduling mechanism that keeps updating the round-trip propagation delay estimate based on real-time ACK/NACK feedback, replacing the one-time estimation used in the original ARQ_HMAC protocol.

2. Implement both the original ARQ_HMAC protocol and the proposed adaptive version in the UnetStack simulation framework.

3. Compare the two approaches using Packet Delivery Ratio (PDR), throughput, number of retransmissions, collision rate, and end-to-end delay under varying traffic loads and changing channel conditions.

1.6 Organization of the Report

The organization of the rest of the paper is as follows. Section 2 is devoted to related work which includes an introduction to the work chronologically in Section 2.1, and an examination of representative algorithms in Section 2.2. The comparison of the above methods is discussed in Section 2.3. Section 3 introduces the proposed adaptive retransmission scheduling scheme, consisting of architecture in Section 3.1, algorithm in Section 3.2, and rationale in Section 3.3.

2 Literature Review

This section walks through the main MAC and ARQ strategies that have been proposed for underwater acoustic sensor networks, starting with a chronological view and then taking a closer look at representative algorithms.

2.1 Existing Approaches in Chronological Order

Early Works (2005–2009): Akyildiz et al. [2] published one of the earliest in-depth analyses of research challenges in UASNs, pointing out the key differences between terrestrial and underwater acoustic communication. This work laid the groundwork for later MAC designs. The Slotted FAMA protocol was introduced by Molins and Stojanovic [8] in 2006, being one of the first handshaking MAC protocols designed for underwater communication networks. In the following year, in 2007, Park and Rodoplu [9] presented their UWAN-MAC protocol. Then in 2009, Molinares and Labrador [7] pointed to the importance of adaptation by designing a logical link layer protocol that adjusted its transmission parameters based on observed channel quality.

Intermediate Developments (2010–2018): Casari and Zorzi [3] have theoretically analyzed the spatial-temporal uncertainty present in underwater channels back in 2011. Several years later, in 2018, Cho et al. [6] studied the asymmetric propagation delay for the case of mobile nodes and suggested a TDMA-based solution for this problem. In the same year, Jiang [4] gave an extensive review of MAC protocols used in underwater communication systems and emphasized that the current trend is towards hybrid solutions.

Recent Developments (2019–2025): The recent developments have been directed towards robust collision avoidance techniques and hybrid solutions. In 2022, Kulla et al. [5] presented another review of the existing MAC protocols paying attention to the trade-offs between contention-based and contention-free protocols. Finally, Dao and Tran [1] in 2025 have developed a new MAC protocol called ARQ_HMAC which serves as a basis for this project. It successfully integrates CSMA-based contention with collision-free retransmission using TDMA but uses static delay estimates.

2.2 Representative Algorithms, with Pros and Cons

This section takes a closer look at four pivotal algorithms that reflect the evolution toward hybrid retransmission protocols, and highlights the limitations that motivate the proposed adaptive mechanism.

2.2.1 Slotted FAMA

Multiple Access using Slotted Floor Allocation has an RTS/CTS handshake mechanism that is designed such that the size of the time slots will be equivalent to the maximum propagation delay. In this way, collisions will occur only when exchanging control packets.

Pros: Effectively eliminates data packet collisions and gives fair access to competing nodes.

Cons: The large slot sizes needed for long propagation delays lead to very poor channel utilization.

Also, the protocol cannot adaptively reschedule retransmissions without going through another full handshake, which is expensive.

2.2.2 UWAN-MAC

UWAN-MAC is an energy-efficient protocol in which nodes decide their own transmission schedules using a localized pseudo-random function. It depends on listening to the channel rather than exchanging explicit synchronization messages.

Pros: Very energy-efficient since it minimizes idle listening and avoids global synchronization overhead.

Cons: It does not deal with collisions dynamically. If the pseudo-random schedules of multiple nodes happen to overlap, they can collide repeatedly. It also lacks any adaptive RTT handling for retransmission scheduling.

2.2.3 TDMA Protocol with Asymmetric Propagation Delay

Cho et al. [6] have proposed this TDMA based scheme that operates under conditions where there may be different propagation delay between a sender and receiver depending on their relative movement in a mobile underwater network scenario.

Advantages: Works well for mobility and asymmetry of delays, providing accurate scheduling in dynamic conditions better than standard TDMA.

Disadvantages: The adjustment of delay occurs at certain intervals, which means that an unexpected change in the environment during this interval can lead to slot mismatching.

2.2.4 ARQ_HMAC

ARQ_HMAC combines asynchronous CSMA and synchronous TDMA in its implementation to send messages the first time and resend them in case of collision, respectively. In case of a collision, the sender generates another time slot using the initial estimate of the propagation delay.

Advantages: Provides consistent performance despite high traffic due to solving the cascaded collision problem with purely contending schemes.

Disadvantages: Propagation delay can be calculated only once during initialization. It does not change dynamically according to environmental conditions and mobility of the nodes.

2.3 Comparison of the Approaches

Table 1 places the reviewed schemes side by side to show the shift from pure contention and contention-free methods toward more complex hybrid approaches. The comparison makes it clear

that while hybrid scheduling deals with initial collisions well, there is still a gap when it comes to mechanisms that can continuously adapt retransmission schedules to the highly variable underwater RTT.

Table 1: Comparison of Existing MAC and ARQ Protocols for UASNs

Protocol	**Type**	**Key Mechanism**	**Pros**	**Cons**
Slotted FAMA [8]	Contention-based	RTS/CTS handshake	Low collision rate	Low utilization, high overhead
UWAN-MAC [9]	Contention-free	Pseudo-random schedule	Energy efficient	Persistent collisions at high load
Adaptive LL [7]	Link-layer	Adaptive parameters	Channel-aware	No MAC-layer retransmission
Asym-TDMA [6]	Contention-free	Asymmetric delay TDMA	Handles mobility	Periodic, not continuous
ARQ_HMAC [1]	Hybrid	CSMA + TDMA retrans.	Stable throughput	Static delay estimate

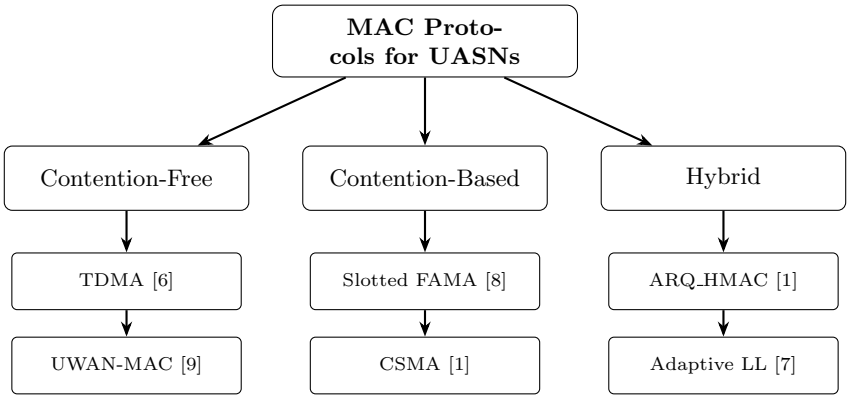


Figure 1: Classification of MAC protocols for underwater acoustic sensor networks.

3 Proposed Approach

This section describes the proposed adaptive retransmission scheduling mechanism, which aims to address the key limitation of ARQ_HMAC identified in the literature review.

3.1 Architecture

Introduction and Motivation: The original ARQ_HMAC scheme [1] requires that all nodes calculate their delay time t_{D_i} from themselves to the master node at initialization, using the first ACK–data exchange. After that, this constant value controls all further TDMA retransmission times. However, in practice, the speed of sound is temperature, salinity, and depth-dependent. Also, with the movement of nodes, the distance between the nodes will be different as well. As a result, the real round-trip time (RTT) will deviate from the initially calculated one and lead to collision in TDMA slots. To solve this problem, a new architecture has been proposed, where an RTT monitor with a change detection module is implemented.

Adaptive RTT Estimation: The system maintains a running estimate $\hat{t}_{D_i}^{(k)}$, which is updated after every successful transmission–acknowledgment exchange using an Exponentially Weighted Moving Average (EWMA) filter:

$$\hat{t}_{D_i}^{(k)} = \alpha \cdot t_{D_i}^{\text{obs},(k)} + (1-\alpha) \cdot \hat{t}_{D_i}^{(k-1)} \tag{1}$$

where $t_{D_i}^{\text{obs},(k)}$ is the observed propagation delay from the k-th successful ACK exchange, computed as:

$$t_{D_i}^{\text{obs},(k)} = \frac{t_{\text{ACK_recv}}^{(k)} - t_{\text{data_send}}^{(k)} - t_d - t_{\text{ACK}}}{2} \tag{2}$$

The parameter $\alpha \in (0,1)$ serves as a smoothing factor. The initial value is taken directly from the ARQ_HMAC network startup phase.

Change Detection Module: To avoid unnecessary processing, the TDMA retransmission schedule is not recomputed with every packet. Instead, a change detection step monitors the absolute difference between the current estimate and the value used when the schedule was last computed ($t_{D_i}^{\text{sched}}$):

$$\Delta_i^{(k)} = \left| \hat{t}_{D_i}^{(k)} - t_{D_i}^{\text{sched}} \right| \tag{3}$$

When $\Delta_i^{(k)}$ exceeds a predefined threshold θ, the node triggers a schedule update by recomputing the TDMA start times and bounds using $\hat{t}_{D_i}^{(k)}$.

3.2 Algorithm

Algorithm 1 shows the step-by-step logic that gets integrated into the node's packet processing workflow, while Figure 2 maps out the logical paths, showing how the system handles successfully and unsuccessfully acknowledged packets.

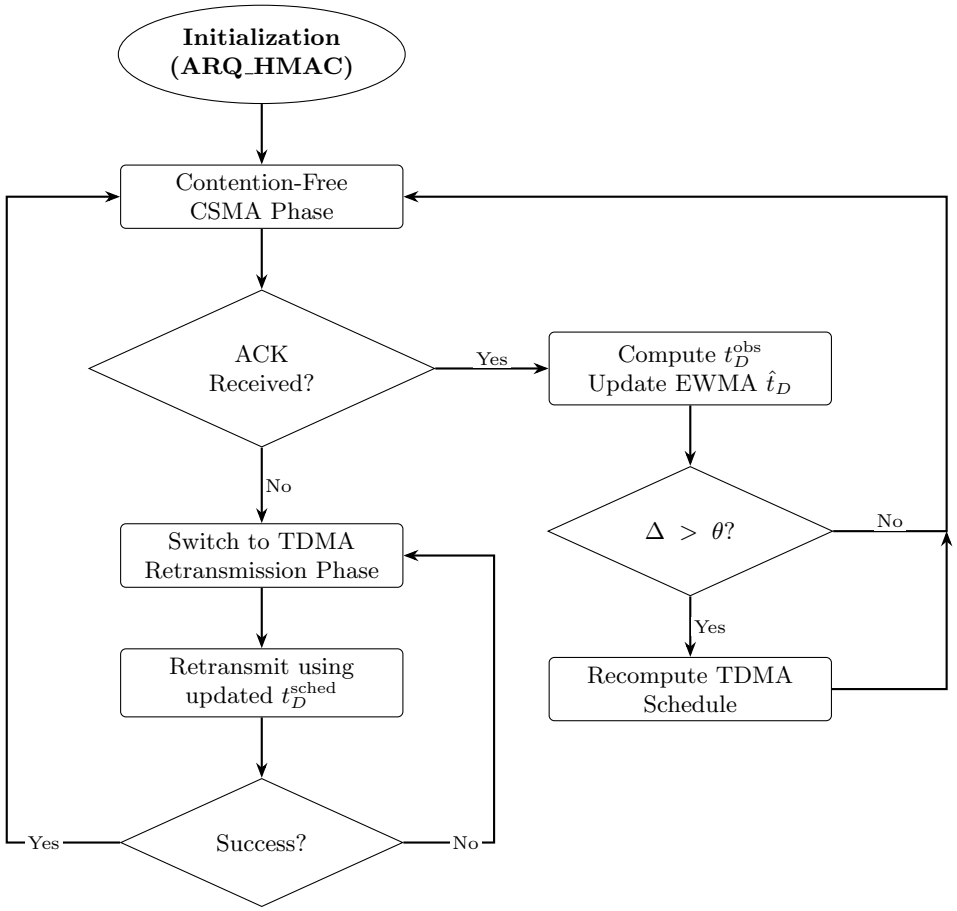


Figure 2: Flowchart of the proposed adaptive retransmission scheduling mechanism integrated into the ARQ_HMAC protocol.

Algorithm 1 Adaptive Retransmission Scheduling in ARQ_HMAC

Require: Initial delay estimate $t_{D_i}^{(0)}$ from ARQ_HMAC initialization, smoothing factor α, threshold θ

Ensure: Updated propagation delay estimate and retransmission schedule

$\hat{t}_{D_i} \leftarrow t_{D_i}^{(0)}$; $\quad t_{D_i}^{\text{sched}} \leftarrow t_{D_i}^{(0)}$
Compute initial TDMA schedule using $t_{D_i}^{\text{sched}}$
for each successful data–ACK exchange k **do**
  Compute $t_{D_i}^{\text{obs},(k)}$ using Eq. (2)
  Update $\hat{t}_{D_i}^{(k)}$ using EWMA (Eq. (1))
  Compute deviation $\Delta_i^{(k)}$ using Eq. (3)
  if $\Delta_i^{(k)} > \theta$ **then**
    $t_{D_i}^{\text{sched}} \leftarrow \hat{t}_{D_i}^{(k)}$
    Recompute $t_{\text{start_TDMA}}$, t_{slot_i}, $t_{\text{time_out_Node}_i}$
  end if
end for
for each transmission failure (NACK / timeout) **do**
  Switch to TDMA retransmission phase
  Use $t_{D_i}^{\text{sched}}$ (updated value) for slot calculation
  Retransmit in computed TDMA slot
end for

3.3 Justification of Design Choices

The design choices behind the proposed adaptive mechanism can be justified as follows:

- **Computational Efficiency:** The EWMA filter needs only a single multiplication and addition per update, which makes it a good fit for resource-constrained underwater nodes where battery life is a serious concern.
- **Keeping Overhead Low:** The change detection threshold θ ensures that the heavier TDMA schedule recomputation only runs when the delay has actually drifted enough to risk slot misalignment.
- **Backward Compatibility:** If the environment happens to stay stable, the EWMA estimate settles quickly, the threshold is never crossed, and the protocol behaves exactly like the original ARQ_HMAC.
- **Directly Addressing the Research Gap:** By shifting the TDMA retransmission slots in real time, the mechanism prevents the kind of cascading collisions that typically hurt underwater protocols during long deployments or when nodes are mobile.

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